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Jingpeng Xie

Hainan University

Lin Zhang

China Agricultural University

Xianli Lu

Hainan University

Ying Fu

Hainan University

Yingxue Guan

Xinyang Agriculture and Forestry University

Huan Liu

Henan University of Science and Technology

Jian Wen

Arcwenjian@gmail.com

Hainan University

Fengqin Cao

Hainan University

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Development and field evaluation of a lure-and-infect strategy integrating attractants with *Beauveria bassiana* for controlling *Bactrocera dorsalis*

Jingpeng Xie¹, Lin Zhang², Xianli Lu¹, Ying Fu¹, Yingxue Guan³, Huan Liu⁴, Jian Wen¹, Fengqin Cao¹

¹School of Tropical Agriculture and Forestry, Hainan University, Haikou 570228, China

²Sanya Research Institute, China Agricultural University, Sanya 572024, China

³School of Foreign Languages, Xinyang Agriculture and Forestry University, Xinyang 464000, China

⁴School of Horticulture and Plant Protection, Henan University of Science and Technology, Luoyang 471023, China

*Corresponding author: Jian Wen (Arcwenjian@gmail.com) and Fengqin Cao (caofengqin@163.com)

Abstract

The wide host range and strong dispersal ability of *Bactrocera dorsalis* (Hendel) have made it a persistent challenge for fruit and vegetable production in tropical and subtropical regions worldwide. Conventional control methods, including attractant-based trapping and the application of entomopathogenic fungi, each have inherent limitations when used alone. This study developed and evaluated a lure-and-infect strategy that integrates attractants with the entomopathogenic fungus *Beauveria bassiana* for the management of *B. bassiana*. First, we optimized a solid-state fermentation medium for *B. bassiana* mass production and identified a peptone-free formulation (C5: rice 171 g, corn meal 20 g, yeast extract 2 g, sucrose 5 g, KH_2PO_4 1 g, and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ 1 g per 200 g substrate) that achieved significantly higher biomass, sporulation, and germination rates compared with other media. Subsequently, compatibility screening revealed that 50% fresh orange juice and 10% β -caryophyllene exhibited minimal inhibitory effects on fungal growth while maintaining attractant activity. Laboratory bioassays demonstrated that the lure-and-infect combinations achieved high mortality within 15 days, with horizontal transmission of conidia to non-exposed individuals. Field trials conducted in a rambutan orchard showed that the *B. bassiana* + 50% orange juice treatment reduced adult density by approximately 80%, increased infection rate, and decreased fruit damage by 40.2 percentage points compared with untreated controls after 75 days. These findings demonstrate that integrating compatible attractants with entomopathogenic fungi can enhance contact probability between pathogens and

highly mobile pests, providing a promising approach for sustainable *B. bassiana* management.

Keywords: Oriental fruit fly; entomopathogenic microorganisms; synergistic effect; food attractant; integrated pest management

Introduction

Among invasive tephritid species, *Bactrocera dorsalis* (Hendel) (Diptera: Tephritidae) ranks as one of the most economically damaging pests, endangering fruit and vegetable production throughout tropical and subtropical regions (Qin et al. 2015; Wan et al. 2011). This species exhibits high reproductive capacity, exceptional flight dispersal ability, and a broad host range encompassing over 400 fruit and vegetable species, enabling rapid population establishment and causing substantial economic losses to agricultural and horticultural industries (Clarke et al. 2005; Stephens et al. 2007; Jin et al. 2020).

Currently, attractant-based control technologies, including food lures and sex pheromones, are widely employed for *B. bassiana* management due to their efficacy and environmental compatibility (Shelly et al. 2014; Haq et al. 2018). The fundamental principle underlying this approach involves attracting adult flies to designated point sources for mass trapping or targeted killing, thereby reducing field population densities. However, the effectiveness of this strategy is highly dependent on the behavioral responses of target individuals. Control efficacy is significantly

compromised for flies that are not successfully attracted, those exhibiting behavioral resistance to attractants, or reproductive females during non-feeding periods (Weldon et al. 2016; Pérez-Staples et al. 2021).

Biological control, particularly the application of entomopathogenic fungi such as *Beauveria bassiana*, provides a promising alternative approach for sustainable *B. bassiana* management (Ekesi et al. 2007; Lacey et al. 2015). *Beauveria bassiana* infects hosts via contact with the cuticle rather than through ingestion, thereby potentially reaching a broader target population (Mascarín and Jaronski 2016). Nevertheless, the high flight activity and mobility of *B. bassiana* result in limited residence time at any single location, greatly reducing opportunities for effective contact with fungal conidia. Consequently, conventional spray applications often show a delayed onset of action and inconsistent field performance (Dimbi et al. 2013; Yousef et al. 2018).

Given the inherent limitations of both attractant-based technologies and entomopathogenic fungi when applied independently, an integrated approach combining these strategies may yield synergistic benefits. The hypothesis underlying this integration is that attractants could efficiently lure *B. bassiana* to pathogen-laden infection devices, achieving dual effects of “attraction” and “sustained pathogenicity” that compensate for the shortcomings of each individual approach. In recent years, the lure-and-infect (also termed attract-and-infect) strategy has shown promising potential for controlling various dipteran pests, including tsetse flies and Mediterranean fruit flies (Maniania 2002; Navarro-Llopis et al. 2013; Dimbi et al. 2013). However,

applying this strategy to *B. bassiana* presents a critical technical challenge: many highly effective attractant components or their complex chemical environments can inhibit conidial germination and mycelial growth of *B. bassiana* (Toledo et al. 2007; Khun et al. 2021). Therefore, screening attractants compatible with *B. bassiana* and optimizing their formulations represent a core prerequisite for establishing such integrated control systems.

Building on this rationale, this study aimed to develop and validate a lure-and-infect synergistic control technology combining specific attractants with *B. bassiana*. Specifically, we first optimized and prepared a solid, cake-shaped substrate carrier suitable for the mass proliferation and sporulation of *B. bassiana*. Subsequently, we systematically evaluated the effects of several commonly used attractants at different concentrations on *B. bassiana* growth within this substrate to identify optimal formulations that combine strong attractiveness with good fungal compatibility. Finally, we conducted field trials in a rambutan (*Nephelium lappaceum*) orchard to assess the practical control efficacy of this integrated strategy compared with single applications of attractants or fungi alone, thereby evaluating its synergistic potential and field applicability. This study provides both theoretical foundations and practical guidance for developing novel, efficient, and environmentally compatible control technologies targeting highly mobile pests.

Materials and methods

Fungal strain and insect rearing

The entomopathogenic fungus used in this study was *Beauveria bassiana* (Bals.-Criv.) Vuill. (strain GDMCC No. 66358). This strain was isolated from adult melon flies (*Zeugodacus cucurbitae* Coquillett) collected in cucumber fields in Wuzhishan City, Hainan Province, China. After purification, the strain was deposited at the Guangdong Microbial Culture Collection Center. Preliminary tests indicated that this strain showed high virulence against tephritid flies under high-temperature tropical conditions. Prior to experiments, the strain was inoculated onto potato dextrose agar (PDA) plates and incubated at 28 °C for 7 days. Fresh conidia were harvested by gently scraping the culture surface and suspended in sterile water containing 0.05% Tween-80 for use in subsequent assays.

Bactrocera dorsalis larvae were originally collected from infested mango fruits at the Haidian Campus of Hainan University, Hainan Province, China. A laboratory colony was established and maintained under controlled conditions for at least 30 generations to ensure a relatively stable genetic background. Larvae were reared on an artificial diet modified from Chang et al. (2006), comprising 5% yeast extract, 10% sucrose, 20% banana pulp, 2% filter paper, 0.1% concentrated hydrochloric acid, 0.2% sodium benzoate, 15% corn starch, and 47.7% distilled water. The adult diet consisted of 3% yeast extract, 10% sucrose, 1% agar, 5% honey, and 81% distilled water (Shelly and Kennelly 2002). All developmental stages were maintained in climate-controlled chambers at 28 ± 1 °C, $70 \pm 5\%$ relative humidity (RH), and a 16L:8D photoperiod. Adults used for behavioral, conidial acquisition, and mortality assays were derived from this laboratory colony, emerged synchronously, starved for 12 h before the

experiments (provided with water only), and were unmated. Except during the starvation phase, adults were continuously supplied with the standard diet and water throughout all indoor attraction assays, population transmission experiments, and mortality tracking assays.

Culture medium optimization

Culture medium optimization experiments were conducted using transparent circular plastic containers (upper diameter 24 cm, lower diameter 20 cm, height 20 cm; Supplementary Fig. S1A) as cultivation vessels, with each container receiving 200 g of total medium as specified in Table 1. The culture media used rice and corn meal as the primary solid substrates, supplemented with sucrose or glucose, yeast extract, peptone, potassium dihydrogen phosphate (KH_2PO_4), and ferrous sulfate heptahydrate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) according to the experimental design. These components and their proportions were based on previous studies, with minor modifications (Wang et al. 2013; Mascarin et al. 2015; Xie 2017; Chen et al. 2024).

Formulations C1 and C2 differed in their soluble carbon source, using sucrose and glucose, respectively. Formulations C3–C7 were derived from C1 by individually removing single components to evaluate the effect of each nutritional component's absence on *B. bassiana* growth and sporulation. To maintain constant total mass, removed portions were replaced with inert diatomaceous earth as a filler. Uninoculated controls corresponding to each formulation were prepared for dry-weight correction.

All raw materials were autoclaved (121 °C, 30 min) and cooled before being placed in cultivation containers. Under aseptic conditions, 10 mL of *B. bassiana* conidial suspension (1×10^8 conidia mL⁻¹) was inoculated and thoroughly mixed. Four biological replicates were prepared for each treatment and incubated at 28 °C and 70–80% relative humidity under dark conditions for 7 days. The method described above produced a white, cake-like mass, which we refer to in this study as the “fungus cake” (Supplementary Fig. S1B). Biomass, conidial yield, and germination rate were determined after incubation.

For biomass determination, samples from each container were dried at 40 °C to constant weight. Dry-weight increment was calculated by subtracting the corresponding uninoculated substrate dry weight, serving as a relative indicator of fungal biomass (Mascarin and Jaronski 2016). For conidial yield determination, three parallel samples (1 g each) were randomly collected from each container, suspended in 10 mL sterile water containing 0.05% Tween-80, shaken vigorously for 30 min, filtered through sterile gauze, and homogenized. Conidia were counted using a hemocytometer under a microscope, and the concentration was expressed as conidia per gram of dry substrate (Oliveira et al. 2015). For germination rate determination, 1 g of dried sample was suspended in 10 mL of 0.1% glucose solution, homogenized, and incubated at 28 °C for 12 h. Two hundred conidia were randomly examined under a microscope, germinated conidia were recorded, and the germination rate was calculated. Each container was assayed in triplicate.

Attractant compatibility screening

Five compounds previously reported to attract *B. bassiana* were selected as candidate attractants: methyl eugenol, linalool oxide, linalool, β -caryophyllene, and fresh orange juice (Tan and Nishida 2012; Jayanthi et al. 2022; Wen et al. 2025). Based on medium optimization results showing that formulation C5 best supported *B. bassiana* growth and sporulation, C5 was used as the base medium to evaluate the effects of different attractants and their concentrations on fungal colony growth, conidial yield, and germination rate.

Methyl eugenol, linalool oxide, linalool, and β -caryophyllene were prepared at three concentrations, 5%, 10%, and 15% (v/v), using 0.05% Tween-80 solution as the solvent. Preliminary experiments indicated that orange juice required concentrations above 30% to elicit significant attraction; therefore, 30%, 50%, and 70% (v/v) were tested. Plates inoculated with fungi and treated with an equivalent volume of 0.05% Tween-80 solution but without attractant addition served as negative controls.

To observe potential inhibitory effects, solid-plate culture was conducted following Khun et al. (2021). C5 components were mixed, ground, and combined with sterile water to form a paste, which was dispensed into sterilized Petri plates and allowed to solidify. Before inoculation, 200 μ L of the attractant solution was uniformly spread across the plate surface and allowed to absorb. Subsequently, a *B. bassiana* agar plug (approximately 2 mm in diameter, 7-day-old culture) was point-inoculated at the center. Plates inoculated with fungi but without attractants served as controls. At least three replicate plates were prepared for each

attractant × concentration combination. Plates were incubated at 28 °C and about 80% RH for 7 days. Colony diameters were measured in two perpendicular directions and averaged.

For conidial yield determination, fungal plugs (6 mm diameter) were excised from uniformly growing colony regions using a sterile cork borer, placed in 5 mL centrifuge tubes containing 3 mL of 0.05% Tween-80 solution, and vortexed for 30 min. The supernatant was collected, and conidia were counted in five randomly selected fields using a hemocytometer to calculate conidial concentration, which was expressed as conidia per unit colony area. For germination rate determination, 6 mm plugs were similarly excised, suspended in 10 mL of 0.1% glucose solution, and incubated at 28 °C for 12 h. Two hundred conidia were randomly examined under a microscope, and germinated conidia were recorded to calculate germination rate. Three replicate plates were prepared for each treatment.

Attraction and conidial acquisition bioassays

Based on attractant screening results, a 10% β -caryophyllene solution and 50% fresh orange juice, which showed good compatibility with *B. bassiana*, were selected for laboratory attraction–infection–acquisition–transmission bioassays.

Infection stations were constructed using transparent plastic boxes (see above) with two opposing openings on the side walls (10 cm long × 5 cm wide) for fly entry and exit. The bottom of each box was lined with approximately 0.3 cm of sterilized cotton. Four treatment categories were established: fungus + attractant, fungus only,

attractant only, and blank control.

For the fungus + attractant treatment, 30 mL of the corresponding attractant solution (10% β -caryophyllene or 50% orange juice) was added to saturate the cotton, followed by placement of a *B. bassiana*-colonized C5 substrate cake (20 cm diameter, cultured at 28 °C for 7 days) to create a lure-and-infect station combining odor release with a fungal inoculum source. For the fungus-only treatment, an identically prepared fungal cake was placed in the box, and the cotton was moistened with 30 mL of 0.05% Tween-80 solution. For the attractant-only treatment, cotton was saturated with 30 mL of the corresponding attractant solution, and an uninoculated sterile substrate cake was placed in the box. For the blank control, cotton was moistened with 30 mL of sterile distilled water only, and an uninoculated sterile substrate cake was placed without any attractant or fungal material.

Behavioral attraction assays were conducted in 100 cm $\times$ 100 cm $\times$ 100 cm mesh cages with treatment devices placed at the cage center. Female and male adults at three developmental stages were tested: 24 h post-emergence (sexually immature), 5 days (pre-mating), and 10 days (pre-oviposition). All test insects were starved for 12 h prior to experiments, with only water provided.

A single fly was released into each cage, and its behavior was continuously observed for 10 min after release. Entry into the box and residence time of at least 5 s were recorded. Individuals entering the box and remaining for ≥ 5 s were considered to show an effective attraction response to the device. Fifty replicates were conducted for each combination of age, sex, treatment, and attractant type. Each adult was used

only once, and fungal cakes were replaced every 10 observations.

Conidial load quantification was conducted using the same devices and conditions as the behavioral attraction assays. Individuals remaining in fungus + attractant devices for at least 5 min were selected and transferred to 1.5 mL microcentrifuge tubes using an aspirator. After brief anesthesia at -20°C , 1 mL of 0.05% Tween-80 solution was added. Tubes were vortexed for 1 min, and the wash solution was examined using a hemocytometer to count conidia. Conidial concentration was calculated and expressed as conidia per individual (Dimbi et al. 2013).

Based on behavioral attraction results, conidial load was determined only for combinations showing significant attraction: the 10% β -caryophyllene treatment showed significant attraction only for 10-day-old females, so only this combination was measured; the 50% orange-juice treatment showed strong attraction for all age groups and both sexes without significant differences, so 10-day-old females were selected as representatives. Fifteen adults were examined for each combination. In addition, 15 age-matched healthy females without fungal-cake contact were selected as blank controls to verify background levels using the same methods.

Population transmission and mortality assays were conducted in $100\text{ cm} \times 100\text{ cm} \times 100\text{ cm}$ mesh cages. Flies were first induced as described above, and individuals remaining in fungus + attractant devices for at least 5 min were designated as “donor” (infected source) individuals. Based on attraction results, 10-day-old females were used for both the 10% β -caryophyllene and the 50% orange

juice treatments. Each replicate consisted of three donor flies, with five replicates in total. Donor flies were marked on one wing with a small waterproof red dot (approximately 2 mm in diameter) for identification.

Three marked individuals were released into each cage together with 30 unmarked *B. bassiana* adults of the same age (15 males and 15 females). The adult diet and water were provided throughout the experiment. Blank controls were established using three age-matched, similarly marked but fungus-unexposed healthy females released with 30 unmarked individuals under identical conditions to assess baseline natural mortality.

At 24 h post-release, 10 unmarked individuals (five males and five females) were randomly captured from each cage. Body-surface conidia were eluted using 0.05% Tween-80 solution and counted following the single-fly conidial-load method to calculate the mean conidial load of unmarked individuals, assessing short-term within-population transmission. Remaining individuals continued to be maintained in the same cages for 15 days, with daily mortality monitoring of unmarked individuals. If marked individuals died within 24 h, they were replaced with new donor flies (or healthy marked flies for blank controls); deaths occurring after 24 h were not replaced, to exclude early mortality due to handling injury while ensuring sufficient transmission time.

All deceased unmarked individuals were transferred to humidified Petri dishes and incubated at 28 °C for 5–7 days. If typical white cottony *B. bassiana* mycelial growth appeared on the cadaver surface, death was attributed to fungal infection.

Fungal infection-derived mortality rates were calculated for each treatment over 15 days.

Field trial

Field trials were conducted from May 12 to August 14, 2024, in a commercial rambutan orchard in the Rezuó Residential Committee, Baoting County, Hainan Province, China, where the mean daily temperature during the experimental period was 28.4 °C. No chemical pesticides or other pest control measures were applied against *B. bassiana* or other pests, and irrigation, fertilization, and pruning practices were consistent across all plots. The orchard covered approximately eight hectares. A randomized complete block design was used, dividing the area into four blocks of about 1.7 ha each, which were further subdivided into six independent plots of approximately 0.2 ha containing at least ten rambutan trees. Six treatments were randomly assigned within each block: (A) fungal cake + 10% β -caryophyllene infection station; (B) fungal cake + 50% orange juice infection station; (C) fungal cake only; (D) 10% β -caryophyllene only; (E) 50% orange juice only; and (F) blank control.

Fungal cakes were prepared as in the laboratory experiments using the C5 solid substrate inoculated with *B. bassiana* and cultured at 28 °C for seven days before deployment, and were placed inside the transparent plastic boxes described above, which had rectangular openings on the side walls. As noted above, the corresponding attractants for each treatment were also placed in the boxes in advance. Treatments A

and B included 10% β -caryophyllene or 50% orange juice solutions absorbed onto sterilized cotton as odor sources. All treatment boxes were suspended within tree canopies using wire at heights of 1.5–1.8 m (Supplementary Fig. S1C-E). To minimize spatial interference, plots within blocks were separated by at least 50 m, with boundaries delineated by roads or landmarks. In each plot, five representative trees were selected according to a five-point sampling pattern, with one box hung per tree (five boxes per plot). Horizontal distance between boxes exceeded 30 cm, and all boxes were replaced every fifteen days.

To monitor treatment effects on field *B. bassiana* populations, six yellow sticky traps (30 × 20 cm) were deployed per plot following Vargas et al. (2008). One trap was placed midway between two edge trees bearing treatment boxes, and two traps were placed approximately 20 m on either side of the central boxes. Traps were deployed one day before trial initiation and again at 15, 30, 45, 60, and 75 days after treatment, exposed for 24 h, then collected and examined for *B. bassiana* adults. Captured flies were transferred to humidified Petri dishes in the laboratory, incubated at 25–28 °C for 5–7 days, and examined under a stereomicroscope for *B. bassiana* mycelial growth to determine field infection levels.

As rambutan fruit began ripening in early August, fruit damage assessment started on day 68 (August 5) of box deployment. Five trees per plot were randomly chosen, and three fruits showing early coloration (green-to-red transition) from each cardinal direction were tagged, totaling 60 marked fruits per plot. On day 75, all tagged fruits were harvested and kept at 28 °C for seven days before dissection. Fruits

containing *B. bassiana* larvae were recorded as infested, and infestation rate was calculated for each treatment plot. Because full fruit maturity coincided with day 75, the orchard manager discontinued deployment of fungal cakes and attractant devices to avoid potential risks to commercial production. Regular integrated pest management practices were resumed, and the field trial concluded on day 75.

Data analysis

All statistical analyses were performed in R (version 4.3.2; R Core Team 2023) using the following packages: *rstatix* (descriptive statistics, nonparametric tests), *lme4* and *lmerTest* (mixed-effects models), *emmeans* (estimated marginal means and multiple comparisons), *car* (analysis of variance), *survival* and *survminer* (survival analysis), and *ggplot2* (data visualization) (Bates et al. 2015; Therneau and Grambsch 2000; Kassambara et al. 2021). Unless specified otherwise, the significance level was $\alpha = 0.05$.

For culture-medium optimization experiments, biomass, conidial yield, and germination rate were analyzed across seven medium formulations (C1–C7). For attractant-compatibility experiments, colony diameter, conidial yield, and germination rate were used to evaluate the effects of different attractants and concentrations on *B. bassiana* growth. Because data frequently violated normality and homoscedasticity assumptions, Kruskal–Wallis tests were applied for overall comparisons. When significant, Dunn’s post hoc tests with Benjamini-Hochberg (BH) correction were used (Dinno 2015).

For laboratory behavioral assays, fly entry into the device (entered = 1, not entered = 0) was treated as a binary response. Separate generalized linear models (GLMs) were built for each attractant (10% β -caryophyllene and 50% orange juice) using a logit link and quasibinomial family to account for overdispersion (Bolker et al. 2009). Fixed effects included treatment (fungus + attractant, fungus only, attractant only, blank control), adult age (1, 5, 10 days), and sex (female, male), plus all interactions. Model simplification followed a stepwise procedure from full three-way models using dispersion estimates from Pearson residuals. Type II deviance analyses with F-statistics tested main and interaction effects; nonsignificant higher-order terms were removed to obtain parsimonious models. Estimated marginal means (EMMs) were computed with emmeans, back-transformed to the probability scale, and compared via Tukey's method (Lenth 2022). Significant interactions were further examined within relevant age $\times$ sex combinations.

Single-fly and population transmission conidial-load data contained zero controls and markedly non-normal distributions. Hence, Wilcoxon rank-sum (Mann–Whitney U) tests compared fungus + attractant treatments with blanks.

For mortality assays, survival time (in days) and status (1 = dead, 0 = alive) were recorded for adults exposed to lure-and-infect stations or blank controls over 15 days. Deaths showing *B. bassiana* mycelial growth were confirmed fungal infections. Survival analyses were performed separately for each attractant using the Kaplan–Meier method, yielding survival curves, probabilities, and medians (95% CIs). Log-rank tests compared curves between fungus + attractant treatments and controls

(Therneau and Grambsch 2000).

For *B. bassiana* population size and *Beauveria bassiana* infection rate data, a linear mixed-effects model (LMM) was used for analysis. Treatment and Day were treated as fixed effects, while Block was included as a random effect to account for variation among blocks. The model was fitted using the `lmer()` function in the *lme4* package. Significance testing was performed using the Satterthwaite or Kenward–Roger method to estimate degrees of freedom, and Type III analyses of variance (ANOVA) were reported. When significant interactions were detected, estimated marginal means (EMMs) were computed to compare treatment-level means averaged across time. LS-means were obtained using the *emmeans* package, and multiple comparison groupings were determined with the *multcomp* package. Multiple comparison correction was conducted using the Tukey–Sidak adjustment ($\alpha = 0.05$). For fruit infestation rates, Kruskal–Wallis tests were used to evaluate overall treatment differences. When significant, Dunn’s tests with Benjamini–Hochberg (BH) correction were subsequently applied.

Results

Culture medium optimization

Medium formulations significantly affected *B. bassiana* biomass ($\chi^2 = 32.2$, $df = 6$, $P < 0.001$), conidial yield ($\chi^2 = 32.8$, $df = 6$, $P < 0.001$), and germination rate ($\chi^2 = 29.2$, $df = 6$, $P < 0.001$) (Fig. 1).

For biomass (Fig. 1A), C5 (peptone-free: rice 171 g, corn meal 20 g, yeast

extract 2 g, sucrose 5 g, KH_2PO_4 1 g, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ 1 g per 200 g) was highest (29.65 ± 1.23 g), followed by C1 (complete formulation: 21.41 ± 0.98 g), C6 (KH_2PO_4 -free: 17.79 ± 0.87 g), C7 ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ -free: 15.82 ± 0.76 g), C3 (yeast extract-free: 11.34 ± 0.71 g), C2 (glucose replacing sucrose: 10.95 ± 0.65 g), and C4 (sucrose-free: 6.07 ± 0.43 g).

For conidial yield (Fig. 1B), C5 and C1 were highest (2.44×10^8 and 2.17×10^8 conidia/mL) without significant difference. C6, C7, and C2 showed intermediate values (4.20×10^6 – 7.02×10^7 conidia/mL), while C3 and C4 were lowest (6.53×10^5 and 2.46×10^5 conidia/mL). Germination (Fig. 1C) showed a similar trend: C5 ($84.51 \pm 1.32\%$) > C6 ($80.11 \pm 1.18\%$) > C1–C3 (73.27 – 76.09%) > C7 ($70.80 \pm 1.05\%$) > C4 ($52.61 \pm 2.14\%$).

Attractant compatibility screening

All tested compounds at different concentrations significantly affected colony diameter (Fig. 2): methyl eugenol ($\chi^2 = 11.0$, $\text{df} = 3$, $P = 0.012$; Fig. 2A), linalool oxide ($\chi^2 = 17.9$, $\text{df} = 3$, $P < 0.001$; Fig. 2B), linalool ($\chi^2 = 10.9$, $\text{df} = 3$, $P = 0.013$; Fig. 2C), β -caryophyllene ($\chi^2 = 16.1$, $\text{df} = 3$, $P = 0.001$; Fig. 2D), and fresh orange juice ($\chi^2 = 15.7$, $\text{df} = 3$, $P = 0.001$; Fig. 2E).

Methyl eugenol and linalool significantly inhibited colony growth at all tested concentrations (5%, 10%, and 15%) compared to controls. Linalool oxide exhibited concentration-dependent inhibition, with stronger effects at higher concentrations. β -Caryophyllene showed the strongest inhibition at 10% concentration, although

differences between 5% and 15% were not significant. Fresh orange juice at 70% concentration resulted in colony diameter comparable to controls, while 30% and 50% concentrations showed significantly smaller colonies, though the reduction was relatively minor. Neither attractant type nor concentration significantly affected conidial yield ($\chi^2 = 0.19\text{--}4.22$, $df = 2$, $P = 0.121\text{--}0.911$; Fig. 3A-E) or germination rate ($\chi^2 = 0.25\text{--}3.68$, $df = 2$, $P = 0.159\text{--}0.882$; Fig. 4A-E).

Attraction and conidial acquisition bioassays

Laboratory bioassays revealed distinct behavioral response patterns between the two attractant systems (Fig. 5A). For the 10% β -caryophyllene treatment, Type II deviance analysis showed that all main effects and interactions were significant (treatment: $F_{3, 216} = 126.7$; age: $F_{2, 216} = 302.0$; sex: $F_{1, 216} = 372.7$; treatment $\times$ age: $F_{6, 216} = 29.7$; treatment $\times$ sex: $F_{3, 216} = 11.2$; age $\times$ sex: $F_{2, 216} = 96.7$; three-way interaction: $F_{6, 216} = 11.6$; all $P < 0.001$). For 1- and 5-day-old adults, attraction rates remained low (2.5–4.0%) across all treatments with no significant differences. However, for 10-day-old females, fungus + attractant and attractant only treatments achieved attraction rates of 36.7% and 41.3%, respectively, significantly higher than fungus only (4.0%) and control (3.6%), while male attraction remained weak ($< 5\%$) regardless of treatment at all ages. In contrast, the 50% fresh orange juice treatment showed only a significant treatment main effect ($F_{3, 216} = 1582.9$, $P < 0.001$), with age ($F_{2, 216} = 0.03$, $P = 0.969$), sex ($F_{1, 216} = 0.11$, $P = 0.744$), and all interactions being non-significant. Fungus + attractant and attractant only treatments achieved attraction

rates of $38.8 \pm 0.65\%$ and $39.1 \pm 0.65\%$, respectively, significantly higher than fungus only and control (3.45–3.58%).

Conidial acquisition (Fig. 5B) and horizontal transmission assays (Fig. 5C) demonstrated that the lure-and-infect system effectively facilitated pathogen transfer. Single-fly measurements showed that individuals exposed to fungus + attractant treatments acquired substantial conidial loads: in the 10% β -caryophyllene treatment, 10-day-old females carried $2,077 \pm 378$ conidia per individual, while in the 50% orange juice treatment, females carried $2,341 \pm 412$ conidia per individual, both significantly higher than controls which showed no detectable conidia ($W = 225$, $P < 0.001$ for both). Population transmission assays further revealed that unmarked individuals acquired conidia through contact with donors within 24 h. In the 10% β -caryophyllene treatment, unmarked individuals carried 56.4 ± 6.55 conidia per individual, while in the 50% orange juice treatment, they carried 68.6 ± 26.4 conidia per individual, both significantly higher than controls ($W = 25$, $P < 0.01$ for both).

Kaplan–Meier survival analysis demonstrated significant treatment effects on mortality. Unmarked individuals in both the fungus + 10% β -caryophyllene (Fig. 5D) and fungus + 50% orange juice (Fig. 5E) treatments showed significantly lower survival probability compared with the control (log-rank: $\chi^2 = 10.27$, $df = 1$, $P = 0.001$ and $\chi^2 = 10.29$, $df = 1$, $P = 0.001$, respectively).

Field trial efficacy

Field efficacy was evaluated through adult population density, fungal infection

rate, and fruit infestation rate over the 75-day monitoring period. Adult abundance was significantly affected by treatment ($F_{5,108} = 109.33$, $P < 0.001$; Fig. 6A), sampling day ($F_{5,108} = 10.94$, $P < 0.001$), and their interaction ($F_{25,108} = 5.53$, $P < 0.001$). Based on overall marginal means averaged across all sampling days, adult abundance was lowest in the fungus + orange juice and fungus + β -caryophyllene treatments (21.6 ± 1.1 and 27.3 ± 1.1 flies trap⁻¹), followed by orange juice only and β -caryophyllene only (34.0 ± 1.1 and 38.6 ± 1.1 flies trap⁻¹), while fungus only and control treatments showed the highest captures (50.9 ± 1.1 and 48.3 ± 1.1 flies trap⁻¹).

Fungal infection rates showed a significant treatment effect ($F_{5,90} = 417.38$, $P < 0.001$; Fig. 6B), whereas sampling day and treatment $\times$ day interaction were not significant. Across all sampling dates, infection rates ranged from 1.7% to 38.3%. The fungus + orange juice treatment produced the highest infection rate ($38.3 \pm 0.7\%$), significantly exceeding all other treatments, followed by fungus + β -caryophyllene ($24.2 \pm 0.7\%$) and fungus only ($8.7 \pm 0.7\%$). The β -caryophyllene only, orange juice only, and control treatments exhibited minimal infection (1.7–3.4%).

Fruit infestation rates also differed significantly among treatments ($\chi^2 = 21.73$, $df = 5$, $P < 0.001$; Fig. 6C). The fungus + 50% orange juice treatment achieved the lowest infestation rate, reducing damage by 40.2 percentage points compared with control, while fungus + 10% β -caryophyllene produced the second lowest rate with a 30.5 percentage point reduction. Neither fungus only nor attractant only treatments differed significantly from control.

Discussion

Bactrocera dorsalis management has long faced two core technical challenges: attractant-based trapping effectively attracts adults but shows limited efficacy against non-responsive individuals or reproductive females during non-feeding periods; entomopathogenic fungi such as *B. bassiana* can infect hosts through cuticular contact independent of feeding behavior, but the highly active flight habits of *B. bassiana* result in insufficient conidia contact, leading to unstable field performance of traditional spray applications. This study established an attract-and-infect synergistic control system through medium optimization, attractant-compatibility screening, and laboratory-to-field validation. Results showed that the peptone-free C5 formulation effectively supported *B. bassiana* growth and sporulation; 50% fresh orange juice and 10% β -caryophyllene maintained strong attractiveness while causing minimal fungal inhibition; and this attract-and-infect method achieved high mortality within 15 days. Field trials using the “fungus + 50% fresh orange juice” treatment reduced pest densities by approximately 80%, with fruit infestation decreasing by 40.2 percentage points compared to controls. These results indicate that recruiting pests to pathogen-enriched stations using attractants may improve contact probability between entomopathogenic fungi and mobile pests, thereby partially compensating for shortcomings of individual control measures.

Effects of medium components on *B. bassiana* growth and sporulation

A noteworthy phenomenon was that the peptone-free C5 formulation

outperformed the complete C1 formulation across all three parameters: biomass, conidial yield, and germination rate. This contradicts conventional expectations, as peptone is generally considered a high-quality nitrogen source supporting fungal growth (Mascarin et al. 2015).

One possible explanation involves the C: N ratio regulating fungal growth-reproduction trade-offs. Previous studies demonstrated that resource competition exists between vegetative growth and sporulation in entomopathogenic fungi, where excessive nitrogen may promote continued mycelial growth while delaying conidial formation. Conversely, moderate nitrogen limitation may serve as a stress signal triggering the transition to reproductive phases (Jaronski and Mascarin 2017). Litwin et al. (2020) similarly observed that nitrogen source type and concentration significantly influence sporulation timing and yield. Therefore, the elevated C:N ratio resulting from peptone removal may more closely approximate optimal conditions for inducing sporulation, though this hypothesis requires verification through systematic C: N ratio gradient experiments.

Yeast extract was confirmed as a critical nutritional factor, as formulation C4 lacking this component showed the poorest performance across all parameters. This likely relates to yeast extract being rich in B-complex vitamins, free amino acids, and trace elements serving as growth factors (Mascarin and Jaronski 2016). The growth inhibition observed when glucose replaced sucrose (C2 formulation) suggests strain-specific carbon source preferences. Sánchez-Rodríguez et al. (2015) also noted that carbon source utilization varies among entomopathogenic fungal strains.

However, this study's medium optimization employed limited component combinations without systematically exploring optimal concentrations or interactive effects, limiting further formulation refinement.

Compatibility between attractants and entomopathogenic fungi

Compatibility between attractants and pathogenic fungi represents a critical prerequisite for successful lure-and-infect strategy implementation and a longstanding challenge (Khun et al. 2021). This study found that different attractant classes exhibited varying effects on *B. bassiana* growth: methyl eugenol and linalool displayed strong antifungal activity at all tested concentrations, while β -caryophyllene and fresh orange juice showed relatively weaker inhibition.

Methyl eugenol, the most widely applied male fruit fly attractant, strongly attracts *B. bassiana* males (Haq et al. 2018), but its phenolic structure may confer antimicrobial properties. Marchese et al. (2017) demonstrated that methyl eugenol exerts antibacterial effects by compromising cell membrane integrity, potentially explaining its inhibition of *B. bassiana*. Linalool, a monoterpene alcohol, has also been reported to possess antifungal activity (Herman et al. 2016). These findings align with Oliveira et al. (2020), suggesting that phenolic and terpene alcohol attractants require careful consideration when combined with entomopathogenic fungi.

In contrast, although β -caryophyllene at 10% concentration inhibited colony diameter, it did not significantly affect conidial yield or germination rate. As a sesquiterpene, β -caryophyllene differs structurally from methyl eugenol and linalool,

lacking hydroxyl groups that may confer stronger antimicrobial activity (Dahham et al. 2015). The transient growth inhibition may stem from nonspecific effects of compound hydrophobicity on membrane permeability, which mycelia may overcome through metabolic adaptation (Alkhaibari et al. 2016). The good compatibility of fresh orange juice may relate to its complex composition: sugars and organic acids may provide additional carbon sources, partially offsetting adverse effects from volatile components (Ekesi et al. 2016).

Nevertheless, this study has methodological limitations. The present valuation focused only on fungal growth, sporulation, and germination, without examining effects on conidial virulence, which are equally important for assessing system efficacy.

Age- and sex-specific behavioral responses

Laboratory behavioral assays revealed distinctly different response patterns between the two attractants: 10% β -caryophyllene was primarily attractive to sexually mature 10-day-old females, while 50% fresh orange juice attracted adults of both sexes across different ages with similar efficacy. These differences may reflect different attractant types acting on different aspects of insect chemical ecology systems.

β -Caryophyllene is a common component of many tropical fruit volatiles and has been reported to exhibit behavioral activity toward *B. bassiana* (Jayanthi et al. 2022). The age- and sex-specific responses observed in this study may relate to the

reproductive physiological status of females. Female *B. bassiana* complete ovarian maturation approximately 7–10 days post-emergence, after which they enter the oviposition phase with heightened sensitivity to host plant volatiles (Pérez-Staples et al. 2021). This sensitivity change may be associated with hormonal alterations and reprogramming of olfactory receptor expression patterns (Liu et al. 2022). β -Caryophyllene may mimic chemical signals released by mature host fruits, thereby selectively activating orientation behavior in females preparing for oviposition. Jaleel et al. (2020) also indicated that behavioral responses of tephritid flies to plant volatiles exhibit marked physiological state dependence.

The broader-spectrum effect of fresh orange juice likely stems from activation of feeding motivation related to basic survival needs. Adults require continuous intake of sugars and proteins to sustain survival and reproduction (Cai et al. 2021), and this nutritional requirement is not constrained by developmental stage or sex. Consequently, fresh orange juice exhibits relatively stable attraction across different physiological states. Piñero et al. (2009) similarly demonstrated that protein- and sugar-based baits attract both sexes of fruit flies.

Conidial transmission and lethal effects

Results indicate that the lure-and-infect system may possess potential for promoting *B. bassiana* conidial acquisition and horizontal transmission. Individuals contacting infection stations for 5 min acquired approximately 2,000 conidia per fly, a load exceeding threshold typically required for effective infection of *B. bassiana*.

Quesada-Moraga et al. (2020) demonstrated that median lethal doses of *B. bassiana* for tephritid flies typically range from 10^2 to 10^4 conidia per individual; therefore, conidial loads observed in this study should be sufficient to cause effective infection.

More ecologically significant were population transmission assay results: infected donors appeared capable of transferring conidia to conspecifics that had not directly contacted infection stations within 24 h, resulting in secondary loads of approximately 60 conidia per individual. Although this secondary load was approximately two orders of magnitude lower than initial contacts, entomopathogenic fungal infections exhibit dose-response relationships, and even lower conidial doses under suitable conditions may still induce sublethal effects or delayed mortality (Fernandes et al. 2015; Souza et al. 2021). Lopes et al. (2020) similarly documented horizontal transmission of entomopathogenic fungi through individual contact within populations, suggesting this transmission pathway may have general applicability.

Survival analysis showed that both lure-and-infect combinations achieved high mortality within 15 days, exceeding natural mortality in control groups. These results suggest that lure-and-infect strategies may enhance control efficacy of fungal agents. However, in natural populations, proportions of individuals directly contacting infection stations may be lower, and transmission efficiency may be influenced by additional environmental factors. Additionally, mechanisms of horizontal transmission (through courtship, grooming, or other behaviors) require further investigation.

Field efficacy and application prospects

Field trial results provided preliminary evidence for practical application potential of the lure-and-infect integrated strategy. Over the 75-day monitoring period, the fungus + 50% orange juice combination reduced yellow sticky trap capture density by approximately 80%, achieved 33.6% fungal infection rate, and reduced fruit infestation by 40.2 percentage points compared to controls. The fungus + 10% β -caryophyllene combination showed slightly lower but still significantly superior efficacy compared to fungal application alone or attractant application alone. These data indicate that integrated application of attractants and *B. bassiana* may produce synergistic effects.

Regarding potential mechanisms underlying this synergy, we speculate several contributing factors. First, attractants may help resolve the "contact bottleneck" problem between pathogens and target pests (Lacey et al. 2015), while lure-and-infect stations actively recruit pests through odor signals, potentially transforming passive contact into active contact. Navarro-Llopis et al. (2013) similarly proposed that attractants can effectively concentrate dispersed pest individuals at limited point sources. Second, *B. bassiana* may provide sustained "pathogenic aftereffect" for attractants. Using attractants alone may reduce adult density through mass trapping but lacks subsequent sustained suppression mechanisms, whereas pathogen introduction not only brings mortality risk to contacted individuals but may also create sustained disease pressure through horizontal transmission (Meyling and Eilenberg 2007). This integrated strategy thus combines dual mechanisms of

behavioral manipulation and biological control, with organic combination of both approaches potentially helping compensate for their individual inherent limitations.

The fungus only treatment showed no significant difference from blank controls in adult density or fruit infestation rate, consistent with our hypothesis that *B. bassiana* field efficacy may be significantly limited without effective contact mechanisms. Jaronski (2010) noted that inconsistent field performance of entomopathogenic fungi largely stems from randomness and inefficiency of conidial-target contact; this study's results appear to support this perspective. Field fungal infection rates remained relatively stable throughout the monitoring period, indicating that the developed fungal cake formulation may possess certain persistence under tropical high-temperature conditions, potentially related to the tropical origin of the strain used (Lacey et al. 2015).

The fungus + 50% orange juice combination outperformed the fungus + 10% β -caryophyllene combination across all evaluation parameters, consistent with laboratory observations of broader target coverage for orange juice. In natural populations containing adults of different developmental stages and both sexes, attractants capable of attracting wider population segments may possess greater application value. Additionally, fresh orange juice is abundant and relatively low-cost, potentially holding practical significance for resource-limited farmers. In practical application, this system may also possess other potential advantages, including relatively minor impacts on non-target organisms, complex fungal action mechanisms that may reduce pest resistance development risk (Wang and Feng 2014), and

potentially good compatibility with other biological control measures (Nikolouli et al. 2018).

However, this study's field trial design has some limitations, and several issues require resolution before large-scale application. Trials were conducted in only one orchard during one growing season; efficacy under different climatic conditions, crop systems, and pest population densities may vary, requiring caution in result extrapolation. The 75-day field trial duration may also be insufficient for comprehensive assessment of long-term effects on pest population dynamics and orchard ecosystems. Furthermore, potential impacts on non-target arthropod communities should be systematically evaluated before widespread implementation.

Future research should address several directions to advance this technology toward practical application. Attractant formulations still have optimization potential, with exploration of fresh orange juice and low-concentration β -caryophyllene combinations warranting consideration. Applicability and economic benefits across different climatic regions and crop systems require larger-scale, multi-location validation trials spanning longer time periods. The horizontal transmission phenomenon observed in this study also suggests that further research on exploiting insect social behaviors to enhance pathogen dispersal may have value, such as screening strains with enhanced host cuticle attachment capability or optimizing infection station design to extend contact duration (Quesada-Moraga et al. 2020).

Conclusion

Through systematic culture medium optimization, attractant compatibility screening, and laboratory and field validation, this study established a lure-and-infect synergistic control system for *B. bassiana*. Results indicate that organically combining specific attractants with *B. bassiana* may enhance contact probability between pathogenic fungi and highly mobile pests, thereby producing synergistic effects. However, given limitations in experimental design and validation scale, these conclusions require verification through additional independent studies. Nevertheless, this research provides references for developing environmentally friendly tephritid pest control technologies, and the established system, following further optimization and validation, may offer a viable solution for fruit and vegetable industry protection in tropical and subtropical regions.

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Table 1 Composition of solid-state fermentation media for different treatment groups (g per 200 g total). C1, complete formulation with sucrose; C2, glucose substituted for sucrose; C3–C7, C1 with individual components removed (yeast extract, sucrose, peptone, KH_2PO_4 , and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, respectively). Removed portions were replaced with diatomaceous earth to maintain constant total mass.

Component	C1	C2	C3	C4	C5	C6	C7
Rice	170	170	172	175	171	171	171
Corn meal	20	20	20	20	20	20	20
Yeast extract	2	2	0	2	2	2	2
Glucose	0	5	0	0	0	0	0
Sucrose	5	0	5	0	5	5	5
Peptone	1	1	1	1	0	1	1
KH_2PO_4	1	1	1	1	1	0	1
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	1	1	1	1	1	1	0

Figure legends

Fig. 1 Optimization of culture medium formulations for *B. bassiana*. Effects of seven different medium formulations (C1–C7) on (A) fungal biomass production, (B) conidial yield, and (C) conidial germination rate. C1: complete formulation; C2: glucose replacing sucrose; C3: yeast extract-free; C4: sucrose-free; C5: peptone-free; C6: KH_2PO_4 -free; C7: $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ -free. Data are presented as mean $\pm$ SE. Different lowercase letters above bars indicate significant differences among treatments (Tukey's HSD, $P < 0.05$).

Fig. 2 Impact of different concentrations of methyl eugenol (A), linalool oxide (B), linalool (C), β -caryophyllene (D), and fresh orange juice (E) on colony diameter. Data are presented as mean $\pm$ SE. The red dashed line indicates the mean value of the control group.

Fig. 3 Impact of different concentrations of methyl eugenol (A), linalool oxide (B), linalool (C), β -caryophyllene (D), and fresh orange juice (E) on conidial yield. Data are presented as mean $\pm$ SE. The red dashed line indicates the mean value of the control group.

Fig. 4 Impact of different concentrations of methyl eugenol (A), linalool oxide (B), linalool (C), β -caryophyllene (D), and fresh orange juice (E) on germination rate. Data are presented as mean $\pm$ SE. The red dashed line indicates the mean value of the control group.

Fig. 5 Behavioral responses and mortality of *B. bassiana* adults exposed to *B. bassiana*–attractant combinations. (A) Attraction rates of adults to 10%

β -caryophyllene-based treatments and 50% fresh orange juice-based treatments. Different lowercase letters above bars indicate significant differences among treatments (Tukey's HSD, $P < 0.05$). (B) Conidial acquisition by single females exposed to Fungus + 10% β -caryophyllene or Fungus + 50% fresh orange juice treatments. (C) Horizontal transmission of conidia to unmarked individuals after 24 h of contact with donors. Values represent the mean number of conidia per individual $\pm$ SE. Different lowercase letters indicate significant differences among treatments (Mann–Whitney U test, $P < 0.01$). Kaplan–Meier survival curves for adults exposed to (D) 10% β -caryophyllene and (E) 50% fresh orange juice treatments. Curves were compared using Log-rank tests ($P < 0.05$).

Fig. 6 Field efficacy of *B. bassiana*-based lure-and-infect systems. Performance of different treatments over a 75-day monitoring period: (A) mean adult abundance (flies trap⁻¹), (B) fungal infection rates (%) of captured flies, and (C) fruit infestation rates (%). Data are expressed as mean $\pm$ SE. Different lowercase letters above symbols or bars indicate significant differences among treatments ($P < 0.05$). For panels A and B, differences were analyzed using linear mixed-effects models; post-hoc comparisons of treatment means (averaged across days) were performed using estimated marginal means (EMMs) with Sidak-adjusted Tukey's tests. For panel C, Tukey's HSD test was used to compare treatments at harvest.

Fig. 1

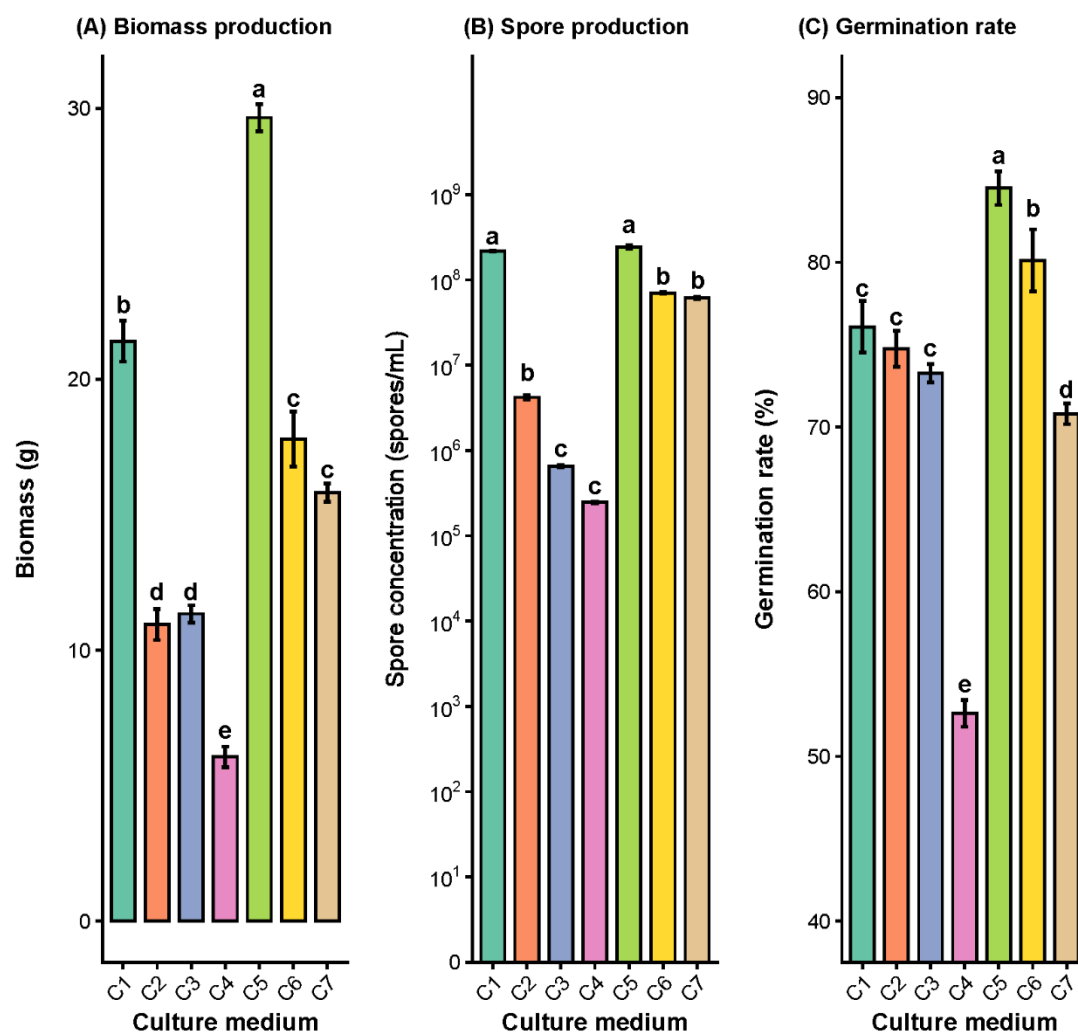


Fig. 2

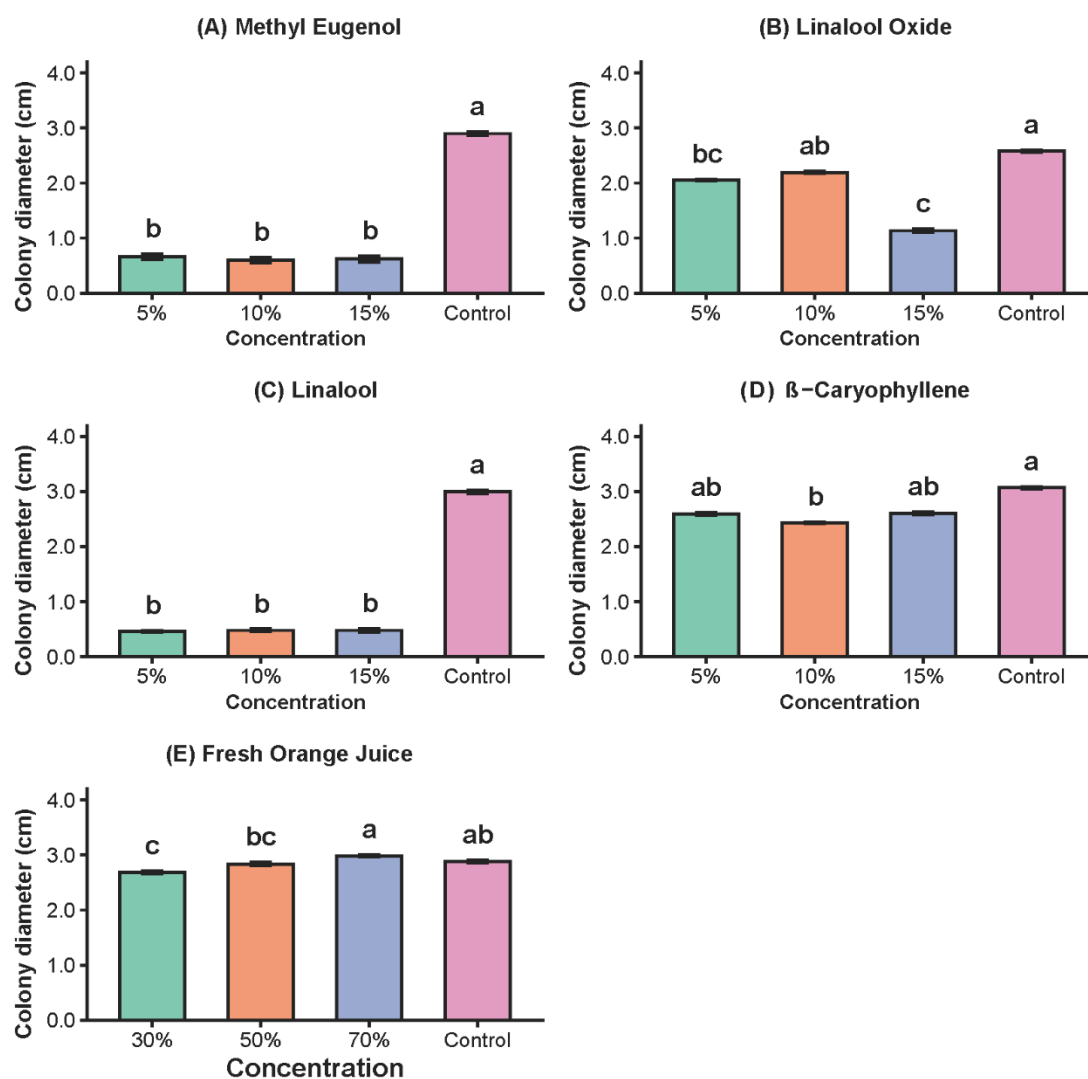


Fig. 3

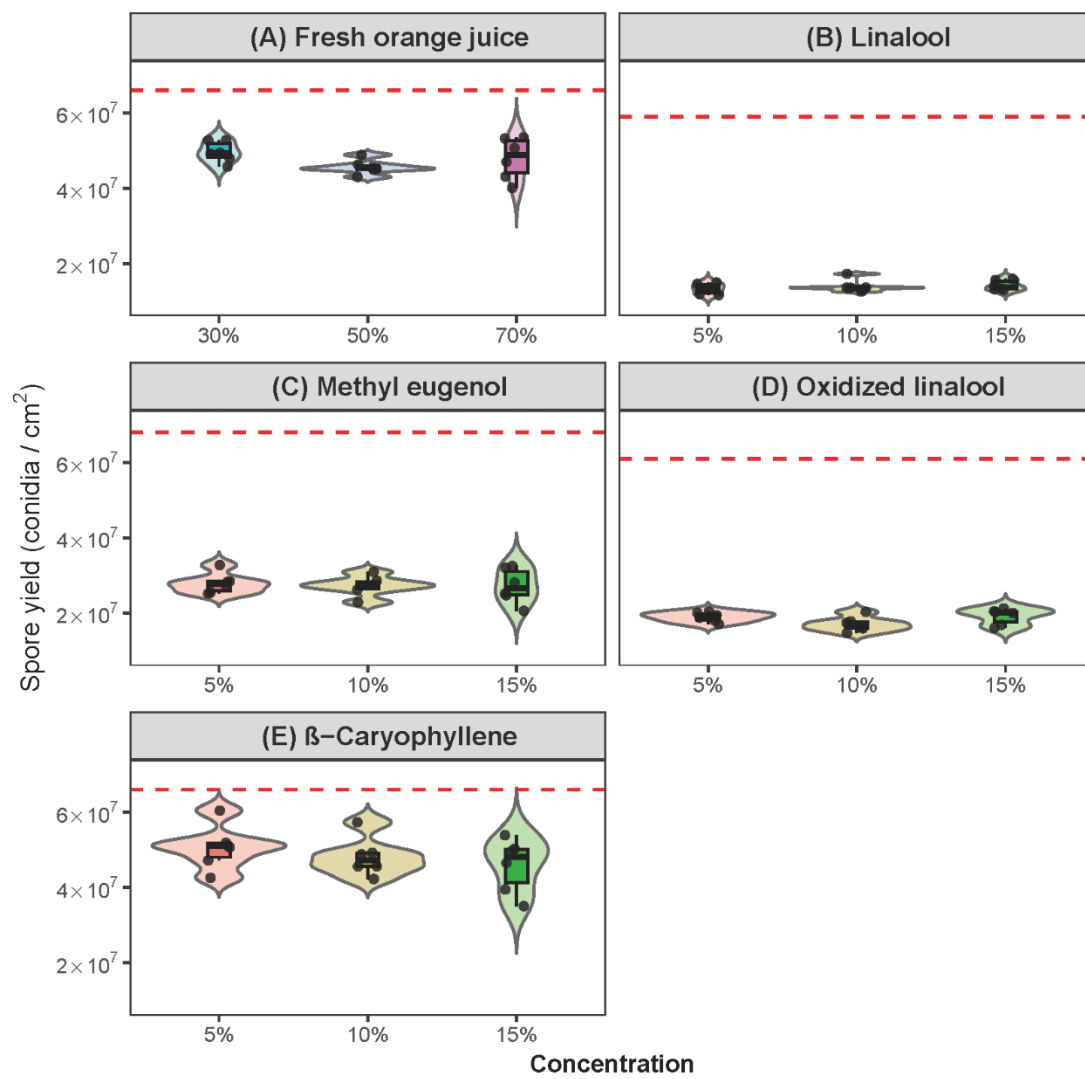


Fig. 4

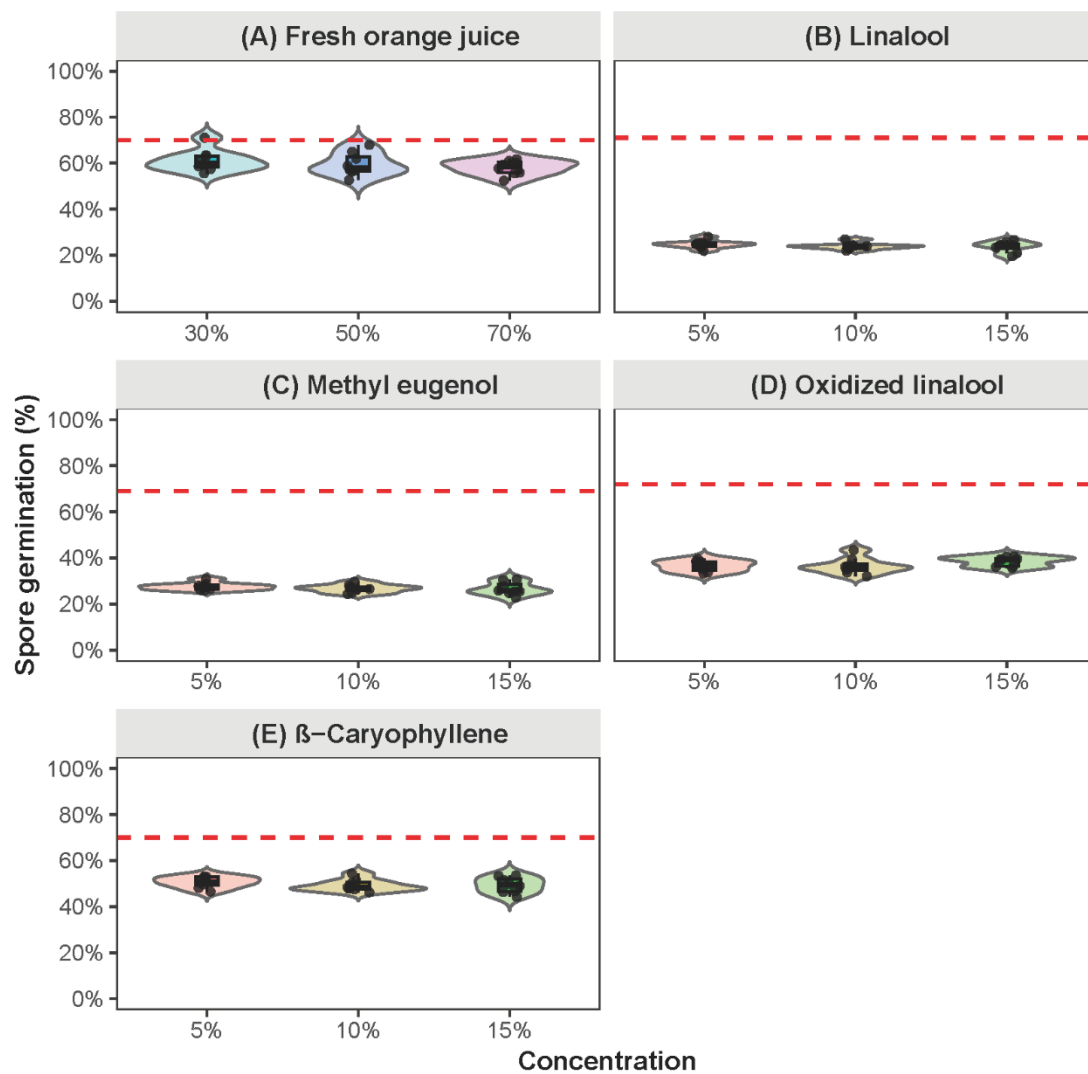


Fig. 5

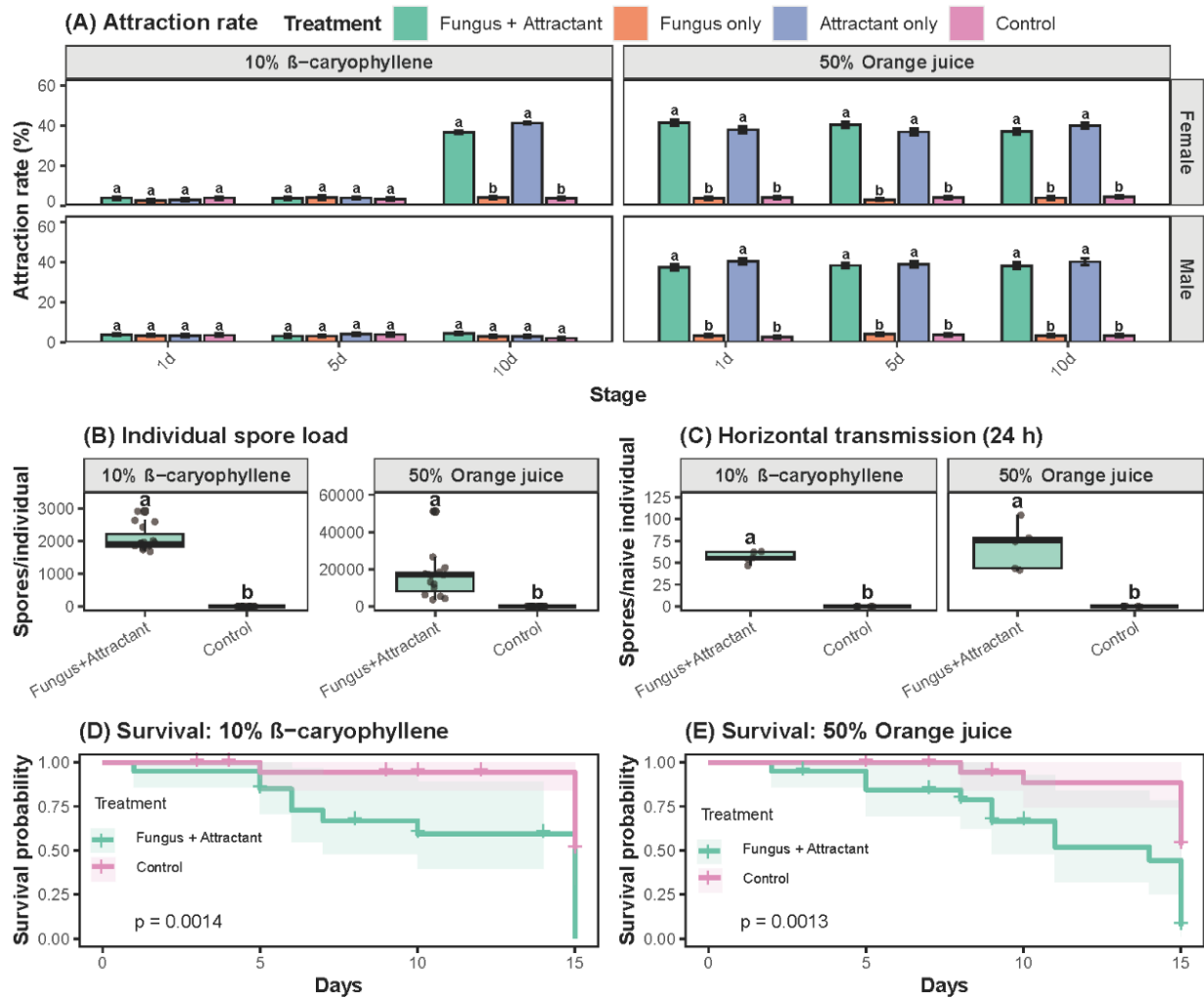
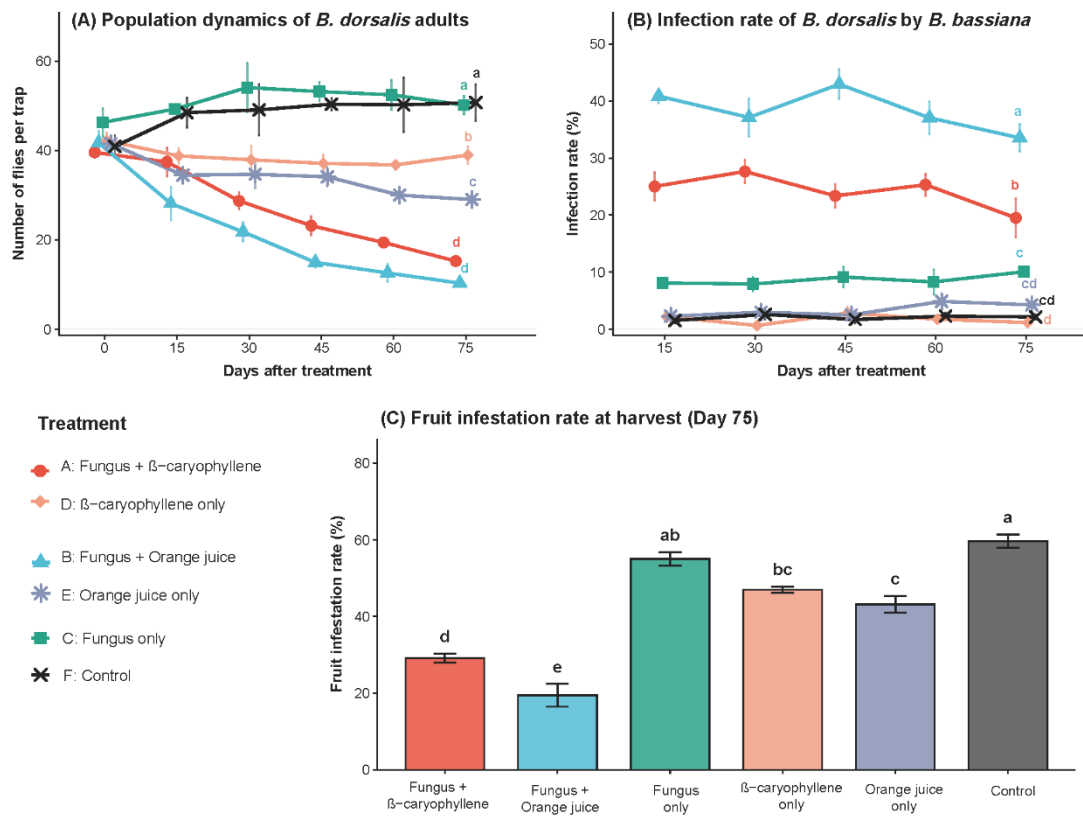


Fig. 6



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